

Siyang Liu

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SUMMARY

PhD candidate in Computer Engineering at the University of Michigan with expertise in large language models, human participant simulation, and web automation. Skilled in building end-to-end systems spanning tokenization, post-training, generative agents, RAG workflows, and web agents. Seeking to advance AI that bridges human-centered design with scalable, real-world impact.

EDUCATION

University of Michigan <i>Doctor of Philosophy in Computer Engineering</i> GPA: 4.00/4.00	Ann Arbor, MI May, 2027
Tsinghua University <i>Master of Computer Science and Technologies</i>	Beijing June, 2021

SKILLS

Large Language Models (LLM): In-context learning, instruction-tuning, preference optimization, RAG, human-in-the-loop post-training.

User Simulation & Agents: Role/agent design, user/participant modeling, multi-agent systems, character synthesis

UI Automation: Web agent framework development (independent), UI automation, full-stack system design

Programming & Systems: Python, Java, JavaScript, C++, concurrency, full-stack development, data processing

RESEARCH EXPERIENCE

University of Michigan / Language and Information Technology Group <i>Graduate Student Research Assistant</i> Advisor: Rada Mihalcea	Ann Arbor, MI 2022-Present
Tsinghua University / Conversational AI Lab <i>Graduate Student Research Assistant</i> Advisor: Minlie Huang	Beijing 2019-2021

WORK EXPERIENCE

University of Michigan/ Language and Information Technology Group <i>Graduate Student Research Assistant</i>	Ann Arbor, MI Present
- Designed a patient-centered RAG system for oncology visit preparation, improving usability (UMUX = 6.0/7), content relevance (6.7/7), and clinical faithfulness (6.82/7); submitted to EMNLP 2025 Demonstration, demo link. - Developed an expert-in-the-loop depression simulation model, combining supervised fine-tuning and preference optimization; produced a realistic 8B LLM outperforming OpenAI baselines in linguistic authenticity and profile adherence (ACL Findings 2025), project link. - Led an age bias study across six LLMs to characterized their bias toward youth values; published at EMNLP 2024. - Engineered task-adaptive tokenizers to enhance long-form generation efficiency by up to 60% while maintaining output quality; published at EMNLP 2023.	
Independent Development <i>Independent developer of an open-sourced project</i>	Ann Arbor, MI August, 2025
- Built WebOasis (github repo), an open agent framework for navigating complex web worlds — “Any site. Any page. Any UI. Any complexity.” - Added one-parameter engine switch (Playwright ↔ Selenium) and a dual-agent architecture for modularity and scalability.	
Educational Testing Service <i>Research Intern</i>	Princeton, NJ June, 2025
- Developed a student-role driven web agent system to simulate digital assessment experiences, accelerating testing and prototyping cycles for educational platforms. - Delivered internship presentation showcasing early progress toward scalable, AI-powered system evaluation.	

Tencent America

Research Intern

Bellevue, WA

May, 2024

- Developed **crowdsourcing framework** treating **AI agents** as recruitable, screenable, and engageable participants across UX research stages; Released open-source toolkit with modular pipeline and **curated billion-scale profile pools** to enable adoption and reuse; Validated at scale in a game prototyping study: 240 simulated players from 2,900 profiles; aggregated agents reproduced ~90% of human findings. One local human $\approx$ 12.8 agents; one crowdsourced human $\approx$ 3.2 agents. Ongoing work.

Kwai (快手)

Algorithm Engineer

Beijing, China

July, 2021

- E-commerce **merchant recommendation with neural graph network**
- Achieved ~5% GMV increase in A/B testing and deployed as a recall strategy on E-commerce merchant recommendation business.

Tsinghua University/Conversational AI Lab

Graduate Student Research Assistant

Beijing, China

July, 2019

- Rectified the widely used **generation diversity metric** Distinct by correcting its mathematical bias toward longer outputs, achieving stronger correlation with human judgment; published at ACL 2022.
- Built **emotional support dialogue systems** and benchmarked the emotional support conversation task, pioneering work in AI for mental health; published at ACL 2021.

PUBLICATIONS

- Liu SY**, Sabour S, Wang X, Mihalcea R, Free Lunch for User Experience: Crowdsourcing Agents for Scalable User Studies, *The Proceedings of the ACM on Human-Computer Interaction (under review)*, 2026
- Liu SY**, An L, Mihalcea R, Patient-Centered RAG for Oncology Visit Aid Following the Ottawa Decision Guide, *arxiv 2507.04026*, 2025
- Liu SY**, Brie B, Li W, Biester L, Lee A, Pennebaker J, Mihalcea R, Eeyore: Realistic Depression Simulation via Expert-in-the-loop Supervised and Preference Optimization, *ACL Findings*, 2025
- Sabour S, Liu JM, **Liu SY**, Yao CZ, Cui SY, Zhang XM, Zhang W, Cao, YR, Bhat A, Guan J, Wu W, Mihalcea R, Althoff T, Lee T, Huang M, Human Decision-making is Susceptible to AI-driven Manipulation, *Nature Communications (under review)*, 2025
- Liu SY**, Maturi T, Shen SQ, Mihalcea R. The Generation Gap: Exploring Age Bias in the Value Systems of Large Language Models. *EMNLP*, 2024.
- Sabour S, **Liu SY**, Zhang ZY, Liu JM, Zhou JF, Suharto A, Li JZ, Lee TMC, Mihalcea R, Huang ML. EmoBench: Evaluating the Emotional Intelligence of Large Language Models. *ACL*, 2024.
- Liu SY**, Deng NH, Sabour S, Huang ML, Mihalcea R. Task-Adaptive Tokenization: Enhancing Long-Form Text Generation Efficacy in Mental Health and Beyond. *EMNLP*, 2023.
- Liu SY***, Sabour S*, Zheng YH, Ke P, Zhu XY, Huang ML. Rethinking and Refining the Distinct Metric. *ACL*, 2022.
- Sun H*, Lin ZR*, Zheng CJ, **Liu SY**, Huang ML. PsyQA: A Chinese Dataset for Generating Long Counseling Text for Mental Health Support. *ACL Findings*, 2021.
- Liu SY***, Zheng CJ*, Demasi O, Li Y, You Z, Huang ML. Towards Emotional Support Dialogue Systems. *ACL* 2021.
- Ke P, Ji HZ, **Liu SY**, Huang ML. Sentilare: Linguistic Knowledge Enhanced Language Representation for Sentiment Analysis. *EMNLP*, 2020.

ACTIVITIES

Reviewer

- Association for Computational Linguistics and ARR Rolling Review, 2019-present.
- Nature Communications, 2024

Organization

- Co-lead for NLP@Michigan Day, 2025
- Host for AI Network at University of Michigan, 2025

Grant Writing Experience

- “AI-assisted Meaning Therapy to Provide Emotional Support for Aging Adults” (2024, not funded)
- “Simulating Depression Experience with Large Language Models” (2025, funded)
- “Medium: People-Centric Knowledge-Controlled Question Generation” (2025, under review)